PROJECT REPORT

IT Help Desk / Ticketing System

1. Project Title

IT Help Desk / Ticketing

2. Student Details

Detail	Information
Student Name	Rajput Karansinh Anopsinh
Enrollment Number	IU2541230537
Division	3CSE-E2
Project Code	CPP62
Project Title	IT Help Desk / Ticketing
Institute	Indus Institute of Technology and Engineering
Program	B.Tech Computer Science & Engineering

3. Problem Statement

In an IT support environment, users may face technical problems related to hardware, software, network connectivity, or other computer-related issues. Managing these problems manually can make it difficult to track requests, assign technicians, record resolutions, and maintain proper records.

The IT Help Desk / Ticketing System provides a simple computerized solution for managing users, support tickets, technicians, and resolutions. It allows help desk staff to create, search, update, delete, process, and store ticket information efficiently.

4. Objectives

The main objectives of the IT Help Desk / Ticketing System are:

- To develop a simple system for managing IT support tickets.
- To maintain records of users, tickets, technicians, and resolutions.
- To provide functions for adding, displaying, searching, updating, and deleting records.

- To process support tickets and record their resolutions.
- To generate reports from the stored ticket information.
- To provide data storage using text files.
- To maintain data persistence by saving and loading records.
- To apply C++ object-oriented programming concepts in a modular project.
- To improve the accuracy and efficiency of IT help desk record management.

5. Features

The system provides the following features:

Category	Feature	Description
Add records	Add User	Adds and stores user information.
	Add Ticket	Creates a new IT support ticket.
	Add Technician	Adds technician information.
	Add Resolution	Records ticket resolution details.
Display records	Display Users	Displays all stored users.
	Display Tickets	Displays all stored tickets.
	Display Technicians	Displays technician records.
	Display Resolutions	Displays resolution records.
Ticket operations	Search Ticket	Searches for a ticket using its details.
	Update Ticket	Updates existing ticket information.
	Delete Ticket	Deletes a selected ticket.
	Process Ticket	Processes a support ticket and handles its resolution.
Reporting	Generate Report	Generates a report of the system records.
Data management	Save Data	Saves records to text files.
	Load Data	Loads previously saved records when the program starts.
	Data Persistence	Saved records remain available after closing and reopening the program.
Reliability	Input Validation	Checks user input and prevents invalid entries.
	Exception Handling	Handles errors during program execution.

6. Classes and Responsibilities

The project is divided into six main classes to keep the program modular and organized.

Class	Responsibility	Key Data Members
User	Stores and manages user information such as user ID, name, email and phone details.	userId, name, email, phone
Ticket	Represents an IT support ticket and stores information related to the reported issue, user and ticket status.	ticketId, userId, technicianId, issue, priority, status
Technician	Stores and manages technician information used for handling IT support tickets.	technicianId, name, specialization
Resolution	Stores resolution information related to processed support tickets.	resolutionId, ticketId, technicianId, solution, date
HelpDesk	Acts as the main system class and manages users, tickets, technicians and resolutions. It provides functions for adding, displaying, searching, updating, deleting and processing records, as well as generating reports.	Arrays of users, tickets, technicians and resolutions, with a count and capacity for each
FileManager	Handles file input/output operations and maintains data persistence by saving and loading users, tickets, technicians and resolutions using text files.	None (provides save and load functions for each record type)

HelpDesk is the central class of the design. It is associated with the four record classes (User, Ticket, Technician and Resolution) and with FileManager, which it uses to save and load data. The complete class diagram is shown in Figure 2.

7. OOP Concepts Used

The project applies the following object-oriented programming concepts:

- Classes and Objects – The system is divided into classes such as User, Ticket, Technician, Resolution, HelpDesk and FileManager, and objects are created to manage their data and operations.

- Encapsulation – Data members and member functions are organized within classes, with appropriate access control.
- Abstraction – The HelpDesk and FileManager classes provide functions that hide the internal implementation details of ticket management and file handling.
- Constructors – Constructors are used to initialize objects of the different classes.
- Modularity – The project is divided into separate header and source files, making the program easier to understand, maintain and modify.
- Exception Handling – Exceptions and input validation are used to handle invalid input and runtime errors.

Inheritance and runtime polymorphism are not used in the current implementation.

8. File Handling

The project uses text files to store and retrieve system data. File handling is implemented through the FileManager class.

The following files are used for data storage:

- users.txt – Stores user records.
- tickets.txt – Stores ticket records.
- technicians.txt – Stores technician records.
- resolutions.txt – Stores resolution records.

The program uses file output operations to save data and file input operations to load previously saved data. The files are located inside the Data folder.

When the program starts, previously saved records can be loaded from the files. The Save Data option stores the current records in the corresponding text files. This provides data persistence, so records remain available even after the program is closed and reopened.

9. UML Diagrams

UML diagrams were created to represent the structure and workflow of the IT Help Desk / Ticketing System. The project includes a Use Case Diagram, Class Diagram and two Sequence Diagrams.
The diagrams help explain the interaction between the help desk staff and the system, the relationships between the classes, and the sequence of operations performed during ticket processing and ticket searching.

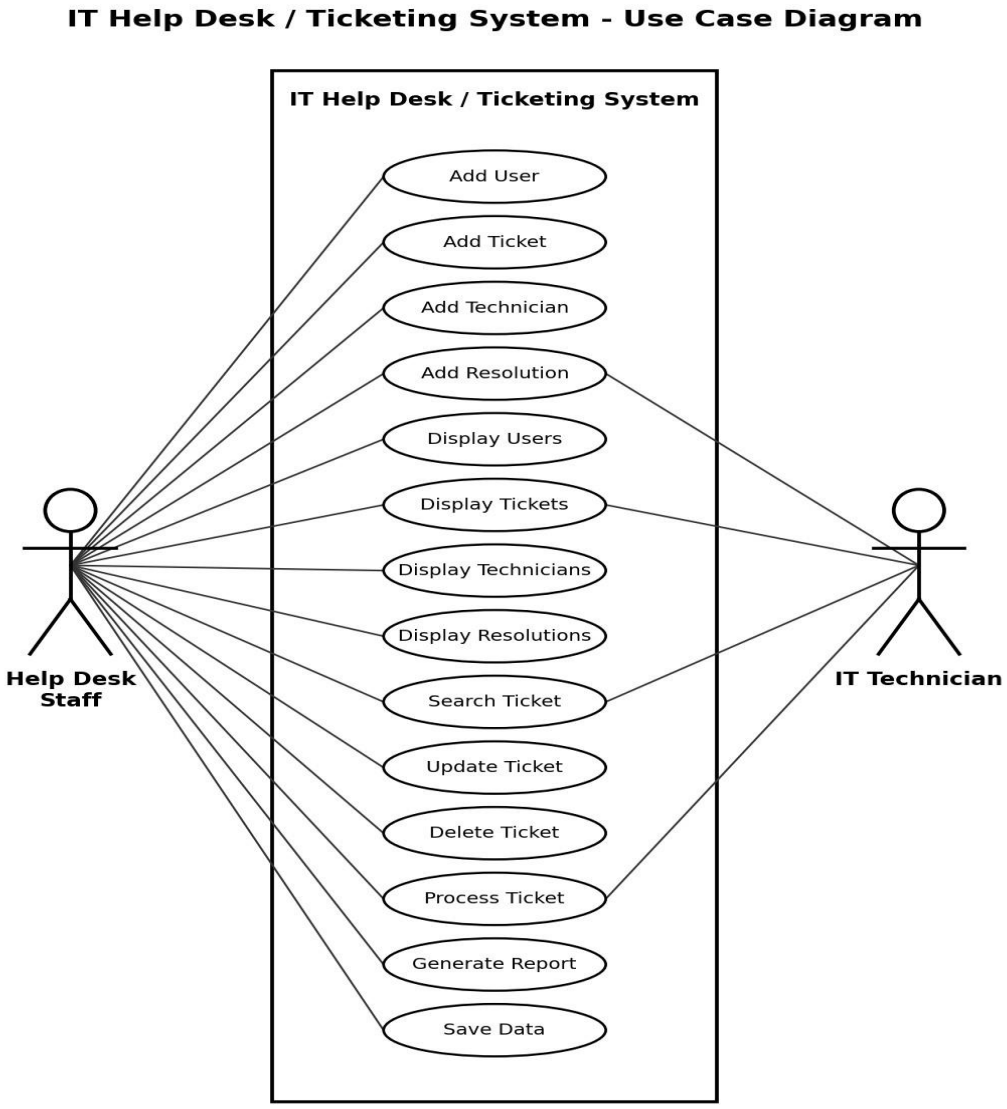


Figure 1: Use Case Diagram

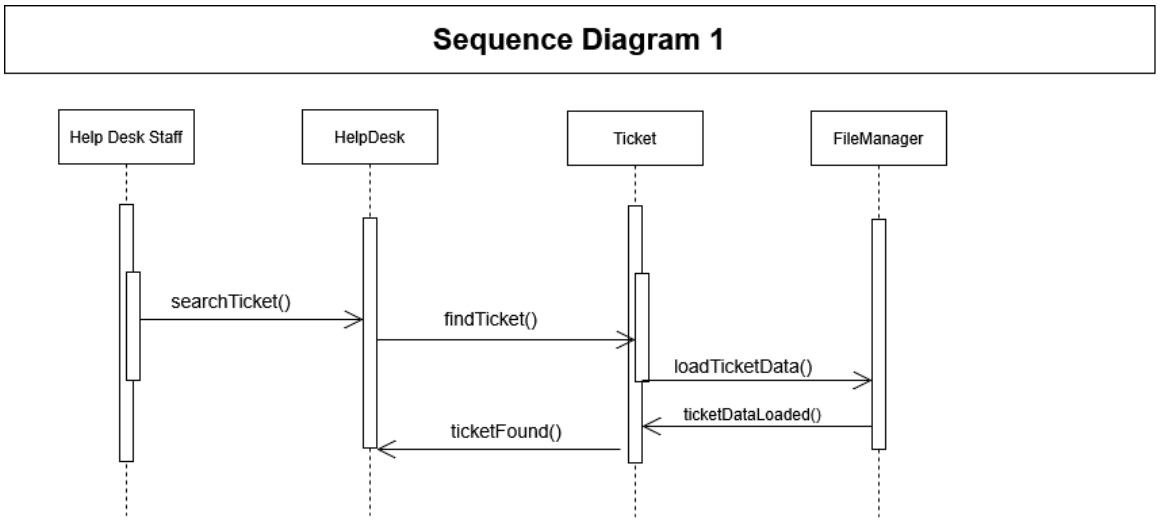


Figure 2: Sequence Diagram 1 – Process Ticket

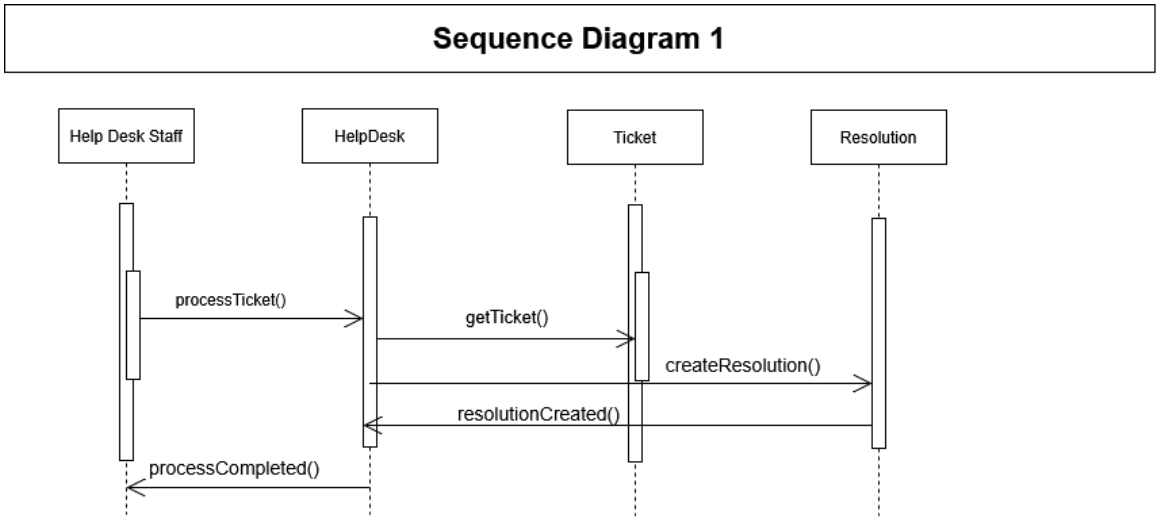


Figure 3: Sequence Diagram 2 – Search Ticket

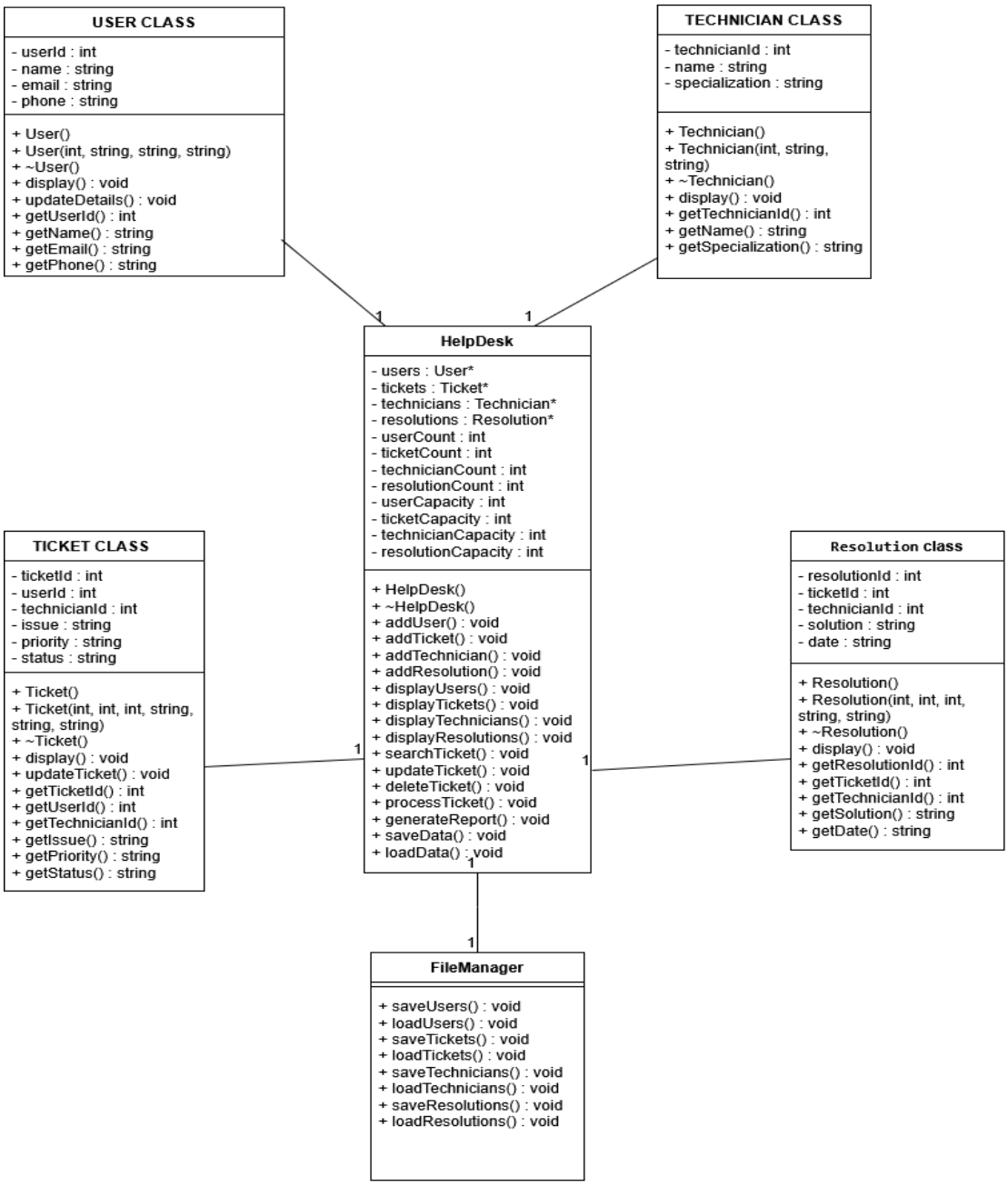


Figure 4: Class Diagram

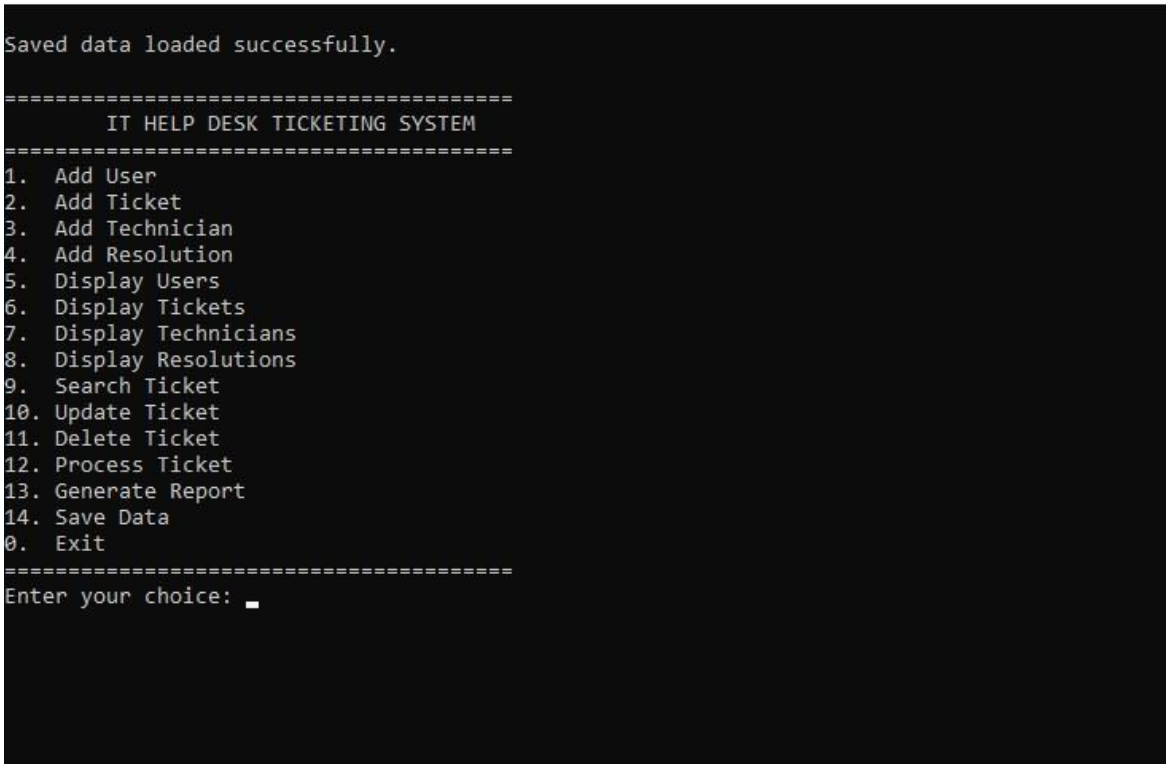
10. Program Workflow

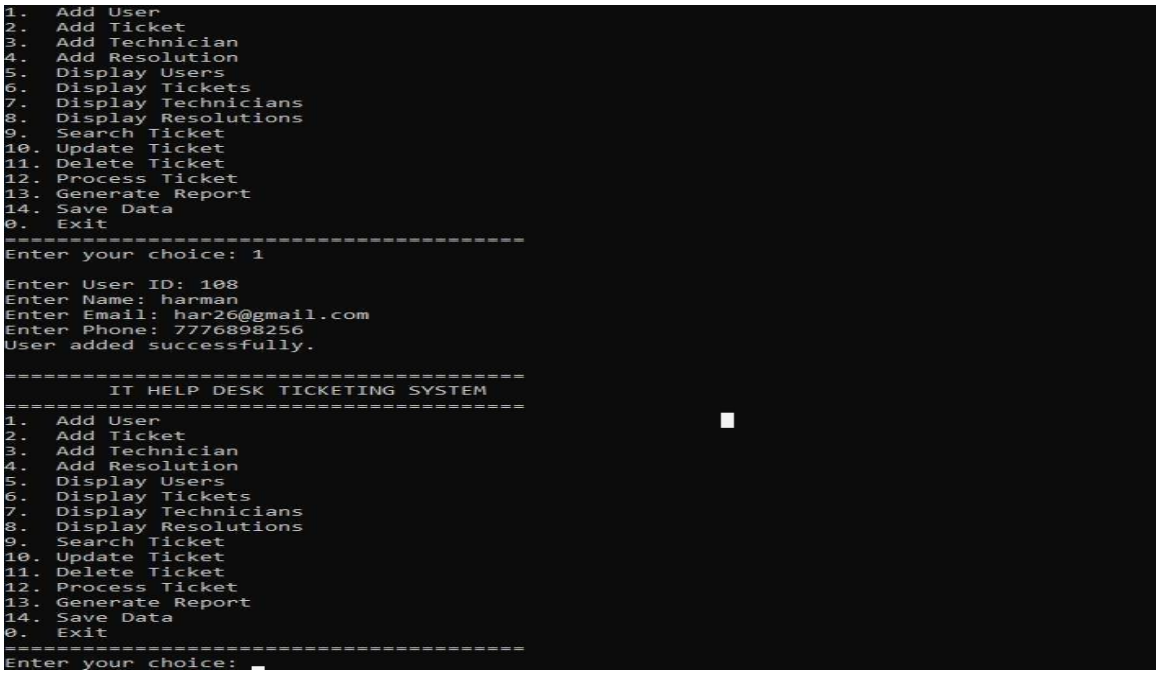
The program follows a menu-driven workflow for managing IT help desk records.

1. The program starts and loads previously saved data from the Data folder.

2. The main menu is displayed to the Help Desk Staff.
3. The user selects an operation from the menu.
4. The selected operation is performed, such as adding, displaying, searching, updating or deleting records.
5. A ticket can be processed and its resolution can be recorded.
6. The system can generate a report of the stored records.
7. The Save Data option stores the current records in text files.
8. The program continues displaying the menu until the user selects the Exit option.
9. On the next program start, previously saved data can be loaded again.

11. Screenshots





```
========================================
1.  Add User
2.  Add Ticket
3.  Add Technician
4.  Add Resolution
5.  Display Users
6.  Display Tickets
7.  Display Technicians
8.  Display Resolutions
9.  Search Ticket
10. Update Ticket
11. Delete Ticket
12. Process Ticket
13. Generate Report
14. Save Data
0.  Exit
========================================
Enter your choice: 3

Enter Technician ID: 301
Enter Name: gaurav
Enter Specialization: Operating system
Technician added successfully.

========================================
```

```
========================================
Enter your choice: 5

===== USER RECORDS =====

--- User Details ---
User ID : 107
Name    : yum
Email   : y12@gmail.com
Phone   : 8734876574

--- User Details ---
User ID : 101
Name    : karan
Email   : karan1@gmail.com
Phone   : 7043589206

--- User Details ---
User ID : 108
Name    : harman
Email   : har26@gmail.com
Phone   : 7776898256

========================================
        IT HELP DESK TICKETING SYSTEM
========================================
1.  Add User
2.  Add Ticket
3.  Add Technician
4.  Add Resolution
5.  Display Users
6.  Display Tickets
7.  Display Technicians
8.  Display Resolutions
9.  Search Ticket
10. Update Ticket
11. Delete Ticket
12. Process Ticket
13. Generate Report
14. Save Data
0.  Exit
========================================
Enter your choice:
```

```
========================================
Enter your choice: 7

===== TECHNICIAN RECORDS =====

--- Technician Details ---
Technician ID : 201
Name          : ram
Specialization: networking

--- Technician Details ---
Technician ID : 301
Name          : gaurav
Specialization: Operating system

========================================
        IT HELP DESK TICKETING SYSTEM
========================================
1.  Add User
2.  Add Ticket
3.  Add Technician
4.  Add Resolution
5.  Display Users
6.  Display Tickets
7.  Display Technicians
8.  Display Resolutions
9.  Search Ticket
10. Update Ticket
11. Delete Ticket
12. Process Ticket
13. Generate Report
14. Save Data
0.  Exit
========================================
Enter your choice:
```

```
5.  Display Users
6.  Display Tickets
7.  Display Technicians
8.  Display Resolutions
9.  Search Ticket
10. Update Ticket
11. Delete Ticket
12. Process Ticket
13. Generate Report
14. Save Data
0.  Exit
========================================
Enter your choice: 6

===== TICKET RECORDS =====

--- Ticket Details ---
Ticket ID      : 1001
User ID        : 101
Issue          : network
Priority       : high
Status         : open
Technician ID  : 201

========================================
        IT HELP DESK TICKETING SYSTEM
========================================
1.  Add User
2.  Add Ticket
3.  Add Technician
4.  Add Resolution
5.  Display Users
6.  Display Tickets
7.  Display Technicians
8.  Display Resolutions
9.  Search Ticket
10. Update Ticket
11. Delete Ticket
12. Process Ticket
13. Generate Report
14. Save Data
0.  Exit
========================================
Enter your choice:
```

```
========================================
Enter your choice: 8

===== RESOLUTION RECORDS =====

--- Resolution Details ---
Resolution ID  : 301
Ticket ID      : 101
Technician ID  : 201
Solution       : connection established
Resolution Date: 01/08/2026

========================================
        IT HELP DESK TICKETING SYSTEM
========================================
1.  Add User
2.  Add Ticket
3.  Add Technician
4.  Add Resolution
5.  Display Users
6.  Display Tickets
7.  Display Technicians
8.  Display Resolutions
9.  Search Ticket
10. Update Ticket
11. Delete Ticket
12. Process Ticket
13. Generate Report
14. Save Data
0.  Exit
========================================
Enter your choice:
```

```
========================================
Enter your choice: 13

========== HELP DESK REPORT ==========
Total Users         : 3
Total Tickets       : 1
Total Technicians   : 2
Total Resolutions   : 1
======================================


========================================
        IT HELP DESK TICKETING SYSTEM
========================================
1.  Add User
2.  Add Ticket
3.  Add Technician
4.  Add Resolution
5.  Display Users
6.  Display Tickets
7.  Display Technicians
8.  Display Resolutions
9.  Search Ticket
10. Update Ticket
11. Delete Ticket
12. Process Ticket
13. Generate Report
14. Save Data
0.  Exit
```

12. Testing

The IT Help Desk / Ticketing System was tested using different operations to verify that the major functions work correctly. Each menu option was tested individually, followed by testing of complete workflows.

Operation	What was verified	Result
Add User	New user records can be added successfully.	Passed
Add Ticket	New support tickets can be created.	Passed
Add Technician	Technician records can be added.	Passed
Add Resolution	Resolution records can be added.	Passed
Display Operations	Stored users, tickets, technicians and resolutions are displayed correctly.	Passed
Search Ticket	Existing tickets can be searched successfully.	Passed
Update Ticket	Ticket information can be updated.	Passed
Delete Ticket	Selected tickets can be deleted.	Passed
Process Ticket	Tickets can be processed and resolution information can be handled.	Passed
Generate Report	The system generates the required report.	Passed
Save Data	Records are saved correctly to text files.	Passed
Load Data	Previously saved records are loaded when the program starts.	Passed
Input Validation	Invalid input is detected and incorrect entries are handled.	Passed
Exception Handling	Error conditions are handled without unexpected termination.	Passed
Data Persistence	Saved records remain available after closing and reopening the program.	Passed

The major functions of the system were tested successfully.

13. Challenges and Solutions

During the development of the IT Help Desk / Ticketing System, several challenges were encountered and resolved.

1. File Handling and Data Persistence
   Challenge: Ensuring that user, ticket, technician and resolution data is saved and loaded correctly.
   Solution: The FileManager class was implemented to handle reading and writing data using text files.
2. Input Validation
   Challenge: Invalid input could cause incorrect program behavior.
   Solution: Input validation and exception handling were added to handle invalid entries safely.
3. Managing Multiple Classes
   Challenge: Managing different types of records in a single program.
   Solution: The project was divided into separate classes and source files to keep the system modular and organized.
4. Ticket Processing
   Challenge: Managing the relationship between tickets and their resolutions.
   Solution: The HelpDesk class coordinates ticket processing and resolution-related operations.
5. Testing and Debugging
   Challenge: Identifying and fixing errors during development.
   Solution: Each menu operation was tested individually, followed by testing of complete workflows an

14. Conclusion

The IT Help Desk / Ticketing System successfully provides a simple and organized solution for managing IT support activities. The system allows help desk staff to manage users, tickets, technicians and resolutions through a menu-driven interface.

The project demonstrates the use of C++ object-oriented programming, modular programming, file handling, input validation and exception handling. The system supports important operations such as adding, displaying, searching, updating, deleting and processing records, along with report generation and data persistence.

The project was tested successfully, including saving and loading data after closing and reopening the program. Overall, the system meets the main objectives of the project and provides a practical foundation for managing IT help desk tickets.

GitHub Repository:

IU2541230537_CPP62_Project_Link